

Figure 2.3 Dope Dyeing

Advantages

- Dyeing is uniform
- Dyed textiles have good to excellent colour fastness to washing and light

Disadvantages

- The strength of solution dyed filaments are slightly lesser
- Expensive method

2.3.1.2 Dyeing at Yarn Stage

The process of dyeing at yarn stage is called as yarn dyeing. Yarn dyeing is preferred to create interesting checks, stripes and plaids in the fabrics. Yarn dyed fabrics are deeper and richer in colour. Fabric with dyed warp and undyed weft are example for chambrays. This includes techniques like skein dyeing, package dyeing, warp-beam dyeing and space dyeing.



Figure 2.4 Yarn Dyeing

Skein Dyeing

In this method, the yarns are loosely wound in hanks or skein form. The hanks or skeins are hung on perforated rods and immersed in a dye bath. The dye solution is circulated in and out of the yarn to achieve an even shade. This method is commonly used for dyeing acrylic and wool yarns.

Advantages

- Loose arrangement of yarn permits excellent dye penetration
- Yarns retain softer feel

Disadvantage

- Most expensive yarn dyeing method

Package Dyeing

Yarns are wound on cones, spools or similar units. These yarn packages are arranged on perforated rods in a rack and immersed in a tank. The dye solution is forced outside from the rods and forced back to the centre under pressure through the packages so that, the dye penetrates the entire yarn.

Advantages

- Dyeing capacity is higher (550 kg) when compared to skein dyeing (200 kg)
- Material to liquor ratio is less
- Uniform dyeing

Disadvantages

- Package dyed yarns do not retain softness and loftiness feel
- Not suitable for high twist yarns which will not allow dye to penetrate

Warp-Beam Dyeing

This method is similar to package dyeing. In warp-beam dyeing, the yarn is wound onto a perforated warp beam and placed in a tank containing dye solution. Dyeing is

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